

Incident Lifecycle Automation in ServiceNow

1. Ideation Phase

1.1 Project Title

Incident Lifecycle Automation in ServiceNow

1.2 Project Overview

The project focuses on automating and standardizing the **Incident Management lifecycle** using ServiceNow. It enables Service Desk agents to create, classify, assign, investigate, resolve, and monitor incidents in a structured manner.

The solution also integrates **Knowledge Management, Change Management, SLA tracking, and Child Incident Management** to improve incident resolution and maintain proper tracking throughout the lifecycle.

1.3 Problem Statement

In many organizations, IT issues such as network failures, VPN problems, hardware issues, and application errors are handled through emails, calls, or informal communication. This can lead to:

- Improper incident classification.
- Delayed assignment to appropriate support teams.
- Difficulty in finding existing solutions.
- Limited visibility into incident status.
- Delayed escalation and resolution.
- Difficulty tracking related changes and incidents.
- Incomplete cause and resolution details.
- SLA compliance issues.

Therefore, a centralized and standardized incident management process is needed.

1.4 Proposed Idea

The proposed solution uses ServiceNow to manage the incident lifecycle through the following process:

Incident Creation → Classification → Knowledge Assistance → Assignment → Investigation → Change/Resolution → Knowledge Creation → Closure

The system supports incident creation, classification, Knowledge Article recommendations, reassignment to Level 2 teams, Emergency Change integration, Child Incident creation, resolution documentation, Knowledge Article creation, and SLA tracking.

1.5 Project Goals

1. The main goals of the project are:
2. Standardize incident creation and classification.
3. Improve incident assignment and escalation.
4. Reduce resolution time using Knowledge Management.
5. Integrate Incident and Change Management.
6. Track related issues using Child Incidents.
7. Maintain proper cause and resolution details.
8. Improve SLA monitoring and service visibility.
9. Provide a complete and traceable incident lifecycle.

Phase 2: Requirement Analysis

2.1 Business Requirements

Requirement	Description
Incident Creation	Service Desk agents should be able to log incidents quickly and consistently.
Incident Classification	Every incident should capture category, subcategory, urgency, service, and configuration item so it's routed correctly from the start.
Knowledge Management Integration	Agents should be able to view and attach relevant knowledge articles to speed up resolution.
Assignment & Reassignment	Incidents should be routable to the correct support group, with the ability to reassign to Level 2 teams when needed.
Escalation & Investigation	Level 2 technicians should have full visibility into incident details, history, and SLA status to investigate and resolve effectively.
Change Integration	Incidents requiring a technical fix should support direct creation and linking of emergency change requests.
Child Incident Tracking	Recurring or related issues should be trackable through child incidents linked to a parent ticket.
Resolution Documentation	Cause and resolution details should be captured before an incident is closed.
Knowledge Article Generation	Resolved incidents should be convertible into knowledge articles for future reuse.
SLA & Compliance Visibility	SLA status should remain visible throughout the incident lifecycle to support compliance tracking and reporting.

2.2 Functional Requirements

FR-01: Service Creation

- **Service:** Remote Access
- **Operational Status:** Operational

FR-02: Service Offering Creation

- **Service Offering:** Corporate VPN
- **Parent Service:** Remote Access
- **Operational Status:** Operational

FR-03: Incident Creation

Service Desk agents should be able to create an incident capturing:

- Short Description
- Caller
- Assignment Group

- Description
- Channel
- Category
- Subcategory
- Urgency
- Service
- Service Offering
- Configuration Item

FR-04: Incident Classification

Incidents should be classified with accurate values:

- **Category:** Network
- **Subcategory:** VPN
- **Urgency:** 2 – Medium
- **Service:** Remote Access
- **Service Offering:** Corporate VPN
- **Configuration Item:** an assigned workstation asset

FR-05: Knowledge Integration

Agent Assist should surface relevant knowledge articles based on the incident description. Agents should be able to:

- View recommended articles
- Open an article
- Mark an article as Helpful
- Attach the article to the incident
- Add comments regarding the article

FR-06: Watch List and Work Notes

The system should allow stakeholders to be added to:

- Watch List
- Work Notes List
- This keeps relevant people informed without transferring ownership of the ticket.

FR-07: Reassignment

Incidents should be re-assignable from Service Desk to the appropriate Level 2 group

Example: **Service Desk** → **Network**

FR-08: Investigation

Level 2 support should be able to:

- View full incident details
- Review knowledge articles
- Review activity history
- View SLA information
- Take ownership of the incident

FR-09: Change Integration

When resolving an incident requires a technical change, the system should support creating and linking an emergency change request. The incident should be placeable in:

- **State:** On Hold
- **Reason:** Awaiting Change

FR-10: Child Incident

The system should allow creation of child incidents for recurring or related issues, linked back to the parent incident.

FR-11: Cause Documentation

The probable cause of the incident should be documented

Example: a suspended service that required a restart.

FR-12: Resolution Documentation

The system should record:

- **Resolution Code:** Workaround provided)
- **Resolution Notes:** service restarted per the linked change request)

FR-13: Knowledge Article Creation

After resolution, the incident should support generating a knowledge article:

- **Knowledge Base:** IT
- **Template:** Standard
- Article should reflect the incident's resolution details

FR-14: SLA Tracking

SLA information should remain visible throughout the incident lifecycle via related records, supporting ongoing compliance tracking.

3. Project Design Phase

3.1 Solution Architecture

The project is designed using ServiceNow's integrated ITSM capabilities to manage the complete incident lifecycle. The main components include **Incident Management**, **Knowledge Management**, **Change Management**, **Configuration Management**, **SLA Management**, and **Child Incident Management**.

High-Level Architecture:



3.2 Service Structure

The service structure used in the project is:

Remote Access



Corporate VPN

- **Remote Access** – Parent Service
- **Corporate VPN** – Service Offering

This service structure provides the required service context for the incident. ServiceNow Incident Management supports associating incidents with services, service offerings, and configuration items.

3.3 Incident Lifecycle Design

The incident is designed to move through a structured lifecycle:

New → In Progress → On Hold → Resolved → Closed

During the lifecycle, the incident can be reassigned to Level 2 support, linked with an Emergency Change, or associated with a Child Incident when required. ServiceNow documents these states as part of its incident lifecycle.

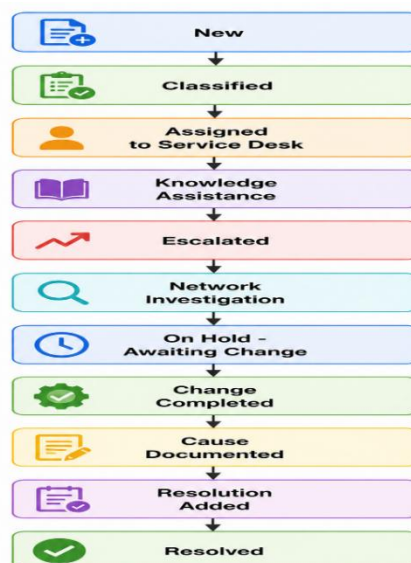


Figure 3.2: Incident Lifecycle Design 1

3.4 Stakeholder Design

Stakeholder	Role	Expected Benefit
End User	Reports IT issues	Faster resolution and better visibility
Service Desk Agent	Creates and classifies incidents	Standardized incident handling
Level 2 Support	Investigates and resolves incidents	Better diagnosis and ownership
Change Team	Handles required emergency changes	Controlled change implementation
ServiceNow Administrator	Maintains the platform	Consistent and manageable process

3.5 Module Integration

The project integrates the following ServiceNow capabilities:

- **Incident Management** – Incident creation and lifecycle management
- **Knowledge Management** – Knowledge-based troubleshooting and article creation
- **Change Management** – Emergency Change integration
- **Configuration Management** – Service and CI information
- **SLA Management** – SLA monitoring and tracking
- **Related Records** – Changes, Child Incidents, SLAs, and Knowledge Articles

This integration provides a connected and traceable incident management process. ServiceNow also supports generating a change from an incident when an IT service change is required.

4. Project Planning Phase

4.1 Project Execution Roadmap

The project is planned to automate and standardize the complete incident management lifecycle in ServiceNow. The project activities are divided into milestones covering service configuration, incident creation, classification, knowledge assistance, escalation, change management, resolution, and knowledge creation.

Milestone	Planned Activity
Milestone 1	Create Service
Milestone 2	Create Service Offering
Milestone 3	Create Incident
Milestone 4	Classify Incident
Milestone 5	Integrate Knowledge
Milestone 6	Update Watch List
Milestone 7	Reassign Incident
Milestone 8	Level 2 Investigation
Milestone 9	Create Emergency Change
Milestone 10	Create Child Incident
Milestone 11	Add Cause
Milestone 12	Add Resolution
Milestone 13	Create Knowledge Article
Milestone 14	Final Validation
Milestone 15	Testing and Deployment

4.2 Task Planning

Phase A – Service Configuration

Objective: Configure the required service and service offering for the incident.

1. Create **Remote Access** service.
2. Set the operational status.
3. Create **Corporate VPN** service offering.
4. Associate the offering with **Remote Access**.

Deliverable: Remote Access service with Corporate VPN service offering.

Phase B – Incident Management

Objective: Create and classify a VPN-related incident.

1. Impersonate **Service Desk** user.
2. Create a **VPN incident**.
3. Add caller and short description.
4. Assign the incident to **Service Desk**.
5. Update incident details.
6. Select **Network** → **VPN** category.
7. Set urgency and priority.
8. Associate service, service offering, and CI.

Deliverable: Properly created and classified VPN incident.

Phase C – Knowledge Management

Objective: Use existing knowledge to assist in resolving the incident.

1. Open **Agent Assist**.
2. Search for relevant VPN Knowledge Articles.
3. Open the appropriate article.
4. Mark the article as **Helpful**.
5. Attach the article to the incident.
6. Add relevant comments.

Deliverable: Relevant Knowledge Article attached to the incident.

Phase D – Collaboration and Escalation

Objective: Collaborate with stakeholders and escalate the incident to the appropriate support team.

1. Add stakeholder to the **Watch List**.
2. Add user to the **Work Notes List**.
3. Reassign the incident to **Network Support**.
4. Verify the **Assigned To** field.
5. Impersonate **Level 2 Technician**.
6. Assign the incident to the technician.
7. Verify incident and SLA information.

Deliverable: Incident successfully escalated to Level 2 support.

Phase E – Change and Child Incident Management

Objective: Manage additional work required to resolve the incident.

1. Update the **Configuration Item**.
2. Set the incident to **On Hold**.
3. Select **Awaiting Change**.
4. Create and link an **Emergency Change**.
5. Create a **Child Incident**.
6. Verify the parent-child relationship.

Deliverable: Emergency change and child incident linked to the parent incident.

Phase F – Resolution

Objective: Document the cause and resolution and complete the incident.

1. Add the **probable cause**.
2. Add the **resolution code**.
3. Add **resolution notes**.
4. Resolve the parent incident.
5. Verify the child incident status.
6. Verify the change request relationship.

Deliverable: Incident resolved with complete resolution information.

Phase G – Knowledge Creation

Objective: Create reusable knowledge based on the incident resolution.

1. Select **Create Knowledge**.
2. Select **IT Knowledge Base**.
3. Select **Standard** template.
4. Verify the article information.
5. Save the Knowledge Article.
6. Verify the article under **Related Records**.

Deliverable: New Knowledge Article created from the incident resolution.

4.3 Project Workflow



4.4 Project Deliverables

Phase	Deliverable
Service Configuration	Remote Access Service and Corporate VPN Offering
Incident Management	Created and classified VPN Incident
Knowledge Management	Knowledge Article attached to Incident
Collaboration & Escalation	Incident assigned to Level 2 Network Support
Change Management	Emergency Change linked to Incident
Child Incident Management	Parent-Child Incident relationship
Resolution	Cause and Resolution details
Knowledge Creation	New IT Knowledge Article
Final Phase	Validated and tested project

4.5 Expected Project Outcome

At the end of the project, the Incident Lifecycle Automation in ServiceNow will demonstrate the complete incident management process from service configuration and incident creation to classification, knowledge assistance, escalation, change management, resolution, and knowledge creation. This ensures that incidents are handled in a structured and traceable manner.

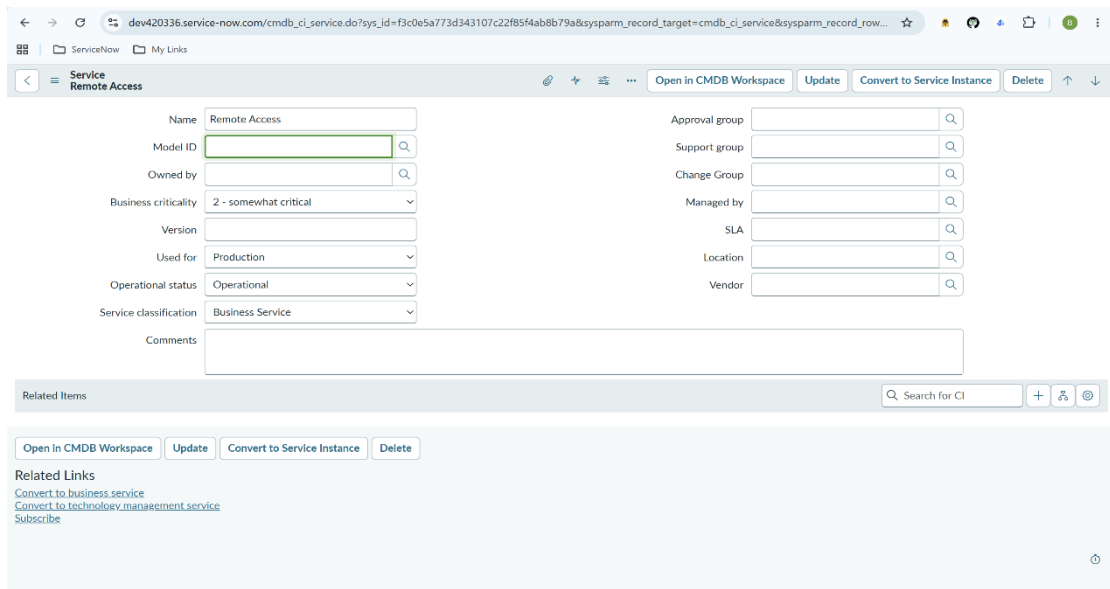
5. Project Development Phase

This phase contains the actual implementation activities performed in ServiceNow.

5.1 Service Configuration

Step 1: Create Service

- Navigate to: **All** → **cmdb_ci_service.list** → **New**
- **Name:** Remote Access
- **Operational Status** : Operational
- Submit the record.

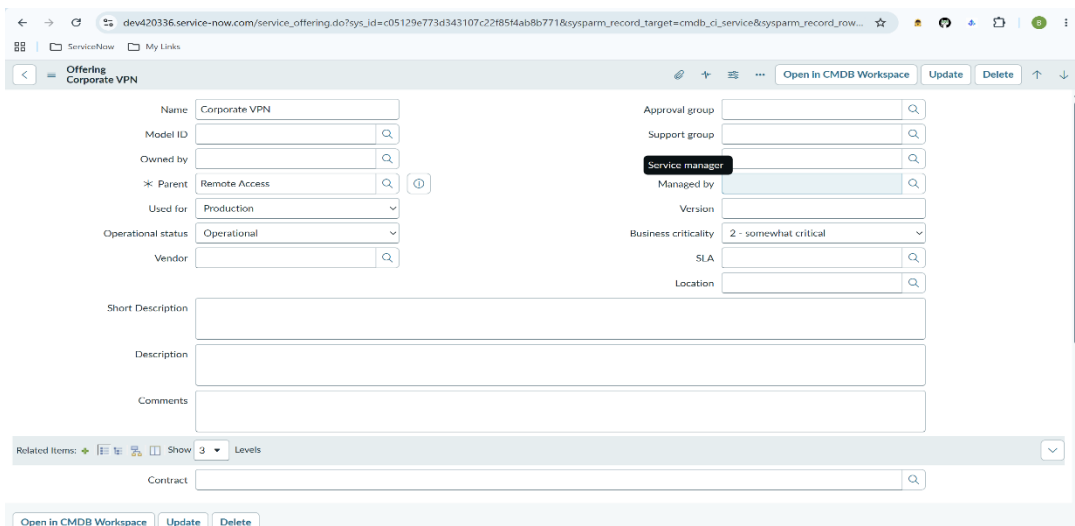


The screenshot shows the 'Service' form in ServiceNow. The browser address bar displays 'dev420336.service-now.com/cmdb_ci_service.do?sys_id=f3c0e5a773d343107c22f85f4ab8b79a&sysparm_record_target=cmdb_ci_service&sysparm_record_row...'. The form is titled 'Service Remote Access'. It includes fields for Name (Remote Access), Model ID, Owned by, Business criticality (2 - somewhat critical), Version, Used for (Production), Operational status (Operational), Service classification (Business Service), and Comments. There are also fields for Approval group, Support group, Change Group, Managed by, SLA, Location, and Vendor. At the bottom, there are buttons for 'Open in CMDB Workspace', 'Update', 'Convert to Service Instance', and 'Delete'. Below the form, there is a 'Related Items' section with a search bar and a 'Related Links' section with links to 'Convert to business service', 'Convert to technology management service', and 'Subscribe'.

Figure 5.1: Service Configuration 1

Step 2: Create Service Offering

- Navigate to: **All** → **service_offering.list** → **New**
- **Name:** Corporate VPN
- **Parent:** Remote Access
- **Operational status:** Operational
- Submit the record.

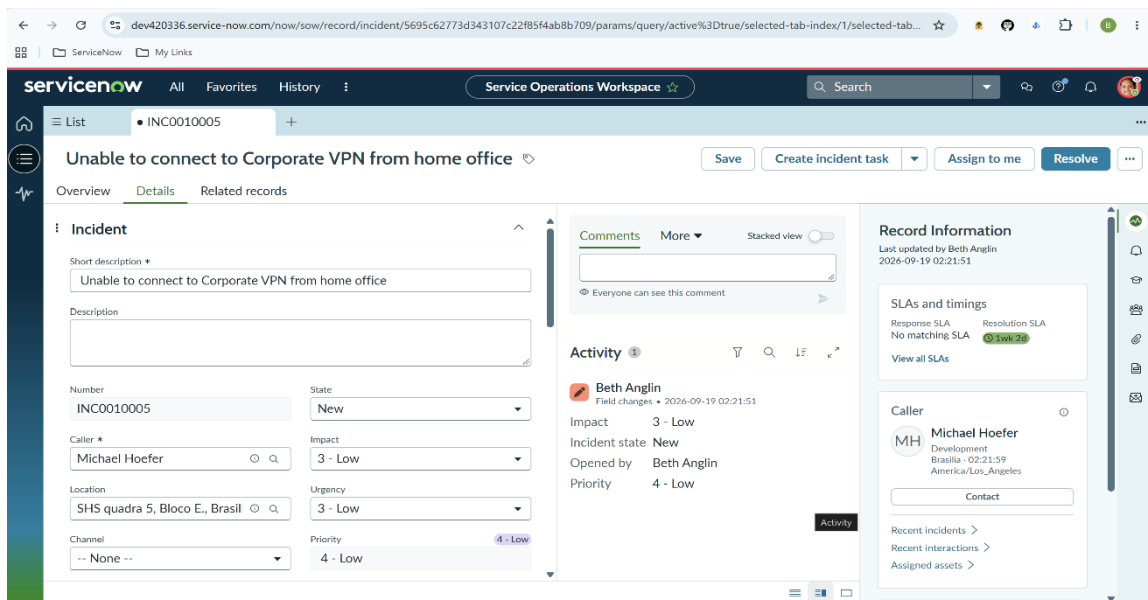


The screenshot shows the 'Service Offering' form in ServiceNow. The browser address bar displays 'dev420336.service-now.com/service_offering.do?sys_id=c05129e773d343107c22f85f4ab8b771&sysparm_record_target=cmdb_ci_service&sysparm_record_row...'. The form is titled 'Offering Corporate VPN'. It includes fields for Name (Corporate VPN), Model ID, Owned by, Parent (Remote Access), Used for (Production), Operational status (Operational), Vendor, Approval group, Support group, Managed by (Service manager), Version, Business criticality (2 - somewhat critical), SLA, Location, Short Description, Description, and Comments. At the bottom, there are buttons for 'Open in CMDB Workspace', 'Update', and 'Delete'. Below the form, there is a 'Related Items' section with a search bar and a 'Contract' field.

Figure 5.2: Service Offering 1

5.2 Incident Creation

- Navigate to: **Service Operations Workspace** → **List** → **Incidents** → **Open** → **New**
- **Short description:** Unable to connect to Corporate VPN from home office
- **Caller:** Michael Hoefer
- **Assignment group:** Service Desk
- Save the incident.



The screenshot shows the ServiceNow 'Incident' creation form in the 'Service Operations Workspace'. The browser address bar shows a URL with an incident ID. The form is titled 'Unable to connect to Corporate VPN from home office'. It includes fields for 'Short description', 'Description', 'Number' (INC0010005), 'State' (New), 'Caller' (Michael Hoefer), 'Location' (SHS quadra 5, Bloco E., Brasil), 'Channel' (None), 'Impact' (3 - Low), 'Urgency' (3 - Low), and 'Priority' (4 - Low). On the right, there is a 'Comments' section, an 'Activity' log showing a field change by Beth Anglin, and a 'Record Information' panel with SLA details and caller information.

Figure 5.3: Incident Creation 1

5.3 Incident Classification

Update the incident with:

- **Description:** Michael advised he was able to connect yesterday evening, but this morning received an authentication error. The Internet works fine.
- **Channel:** Phone
- **Category:** Network
- **Subcategory:** VPN
- **Urgency:** 2 - Medium
- **Service:** Remote Access
- **Service Offering:** Corporate VPN
- **Configuration Item:** ThinkStationS20(Any one)
- **Work notes:** Classified and assigned incident.
- Then select **Assign to me** and save.

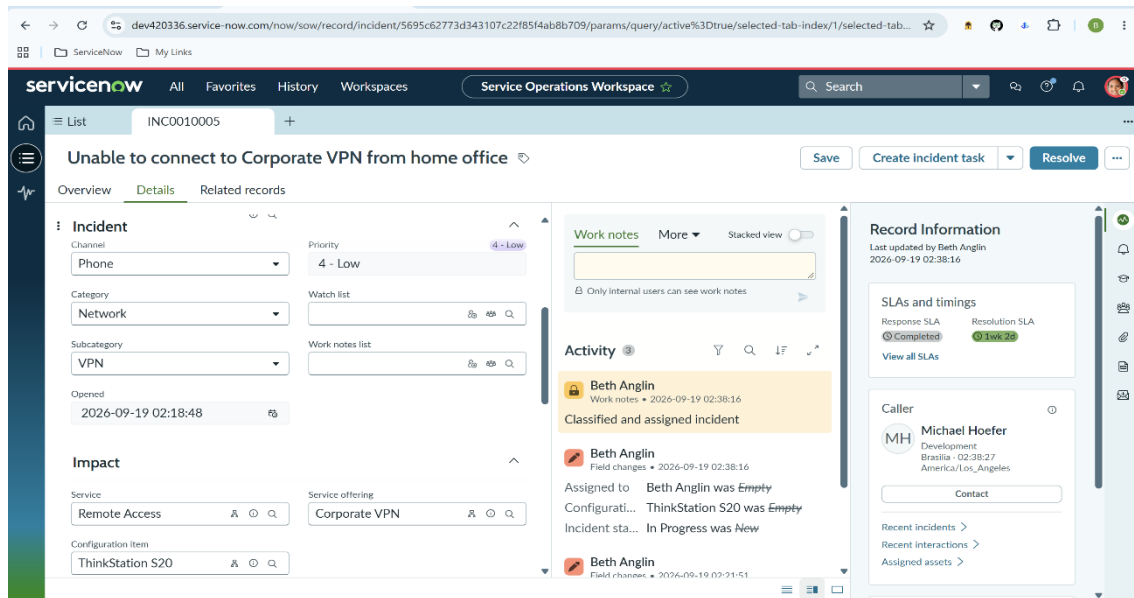


Figure 5.4: Incident Classification 1

5.4 Knowledge Integration (Agent Assist)

- Open **Agent Assist** from the right-side panel.
- Open a relevant Knowledge Article.
- Select the three-dot menu.
- Mark the article as **Helpful**.
- Select **Attach Article**.
- Add the comment: Follow the steps and resolve the incident.
- Attach the article.
- Return to the incident.
- Verify that the article is recorded in the Activity log.

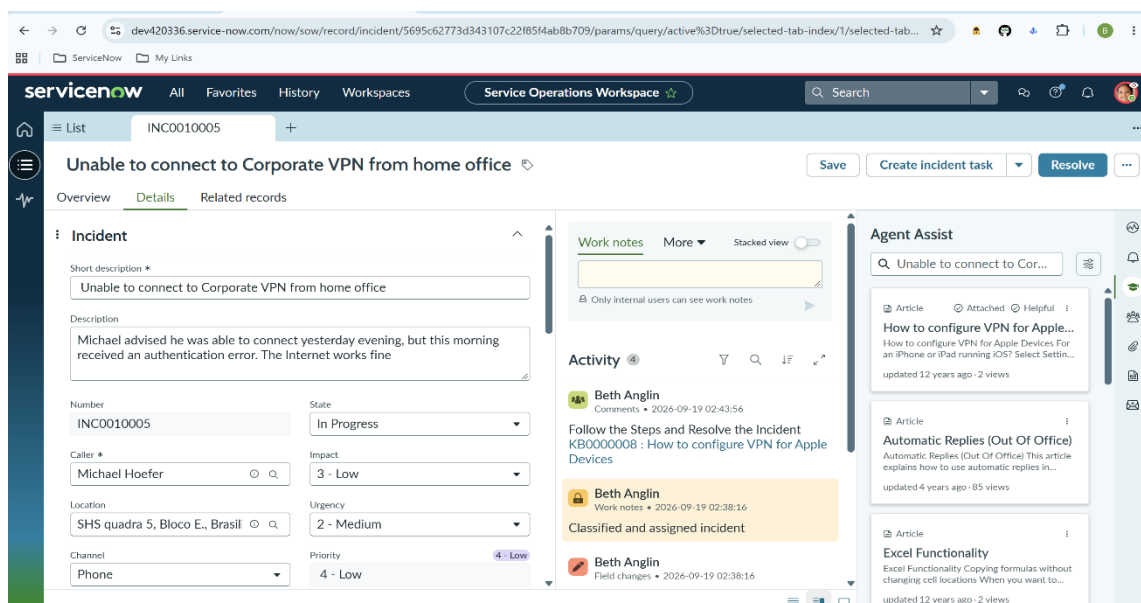


Figure 5.5: Knowledge Integration 1

5.5 Watch List & Work Notes

In the incident **Details** tab:

- Add **Samantha Bordwell** to Watch List.
- Add **Beth Anglin** to Work Notes List.

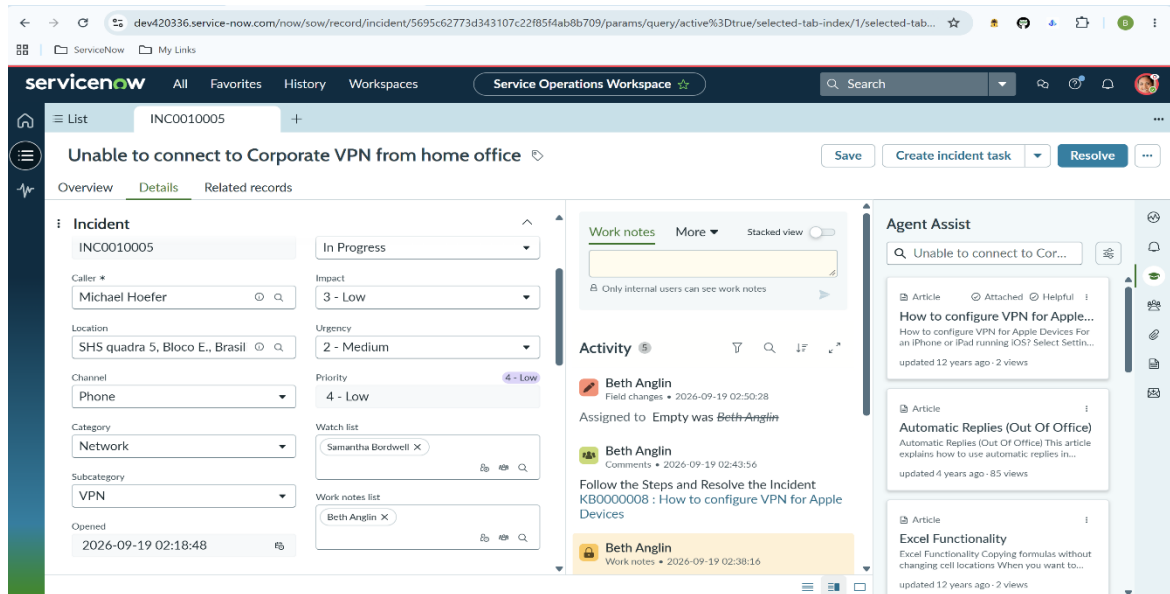


Figure 5.6: Watch List & Work Notes 1

5.6 Incident Reassignment

- Update: **Assignment Group** → **Network**
- Verify that the **Assigned To** field is cleared automatically when the assignment group changes.
- Save the incident.

5.7 Level 2 Investigation

- Impersonate the Level 2 Network technician.
- Navigate to: **Service Operations Workspace** → **Incidents** → **Open**
- Open the VPN incident.
- Select **Assign to me**.
- Verify:
 - Description visible
 - Knowledge Articles are available.
 - SLA visible in Task SLAs in Related records Section

5.8 Emergency Change Integration

- Update Configuration Item from ThickStation to PowerEdge.
- **State:** On Hold
- **On Hold Reason:** Awaiting Change
- Save.

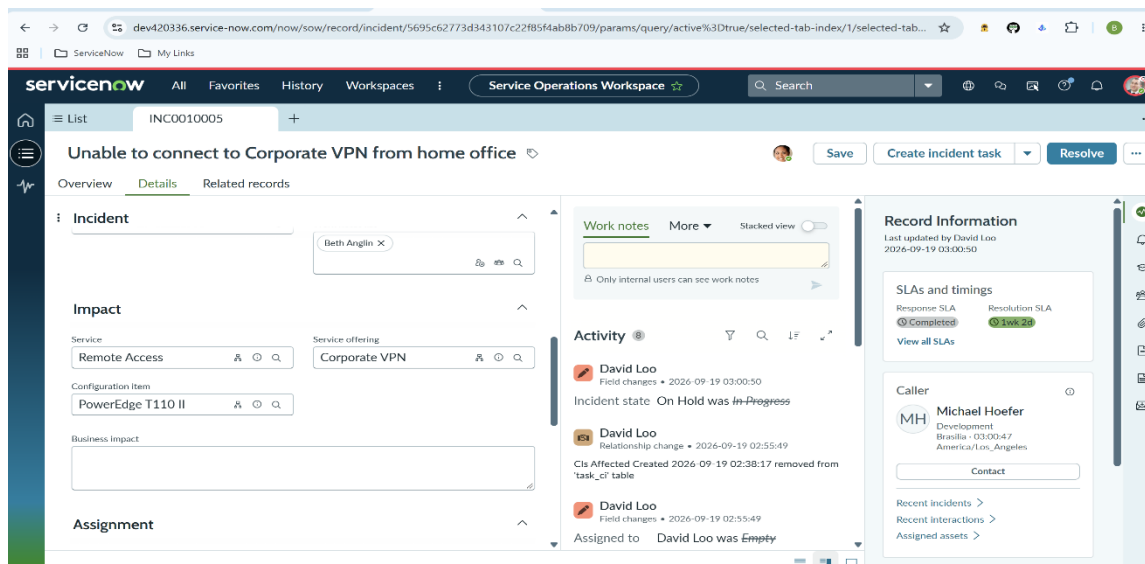


Figure 5.7: Change Integration 1

5.9 Child Incident Creation

- Navigate to: **Related Records** → **Child Incidents** → **New**
- **Short Description:** Unable to connect to Corporate VPN
- **Caller:** Select any user
- **Description:** User receiving VPN authentication failure error.
- Save the record.
- Verify that the child incident appears under **Related Records**.

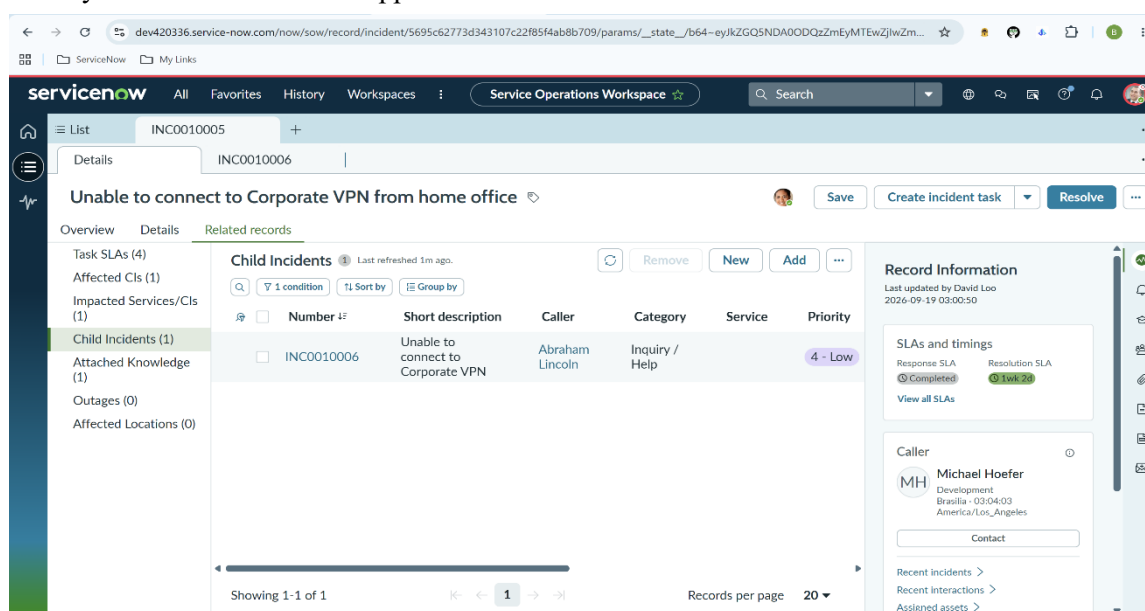


Figure 5.8: Child Incident Creation 1

5.10 Cause Documentation

From the Overview tab, select **Add Cause**.

- **Probable Cause:** PowerEdge service was suspended and required restart.
- Save the cause information.

5.11 Resolution Documentation

Select **Add Resolution**.

- **Resolution Code:** Workaround provided
- **Resolution Notes:** Restarted VPN-SRV-02 service as per emergency change request.
- Save resolution.
- Click **Resolve**
- Confirm popup → Resolve.
- Save the resolution.
- Select **Resolve** and confirm the resolution.

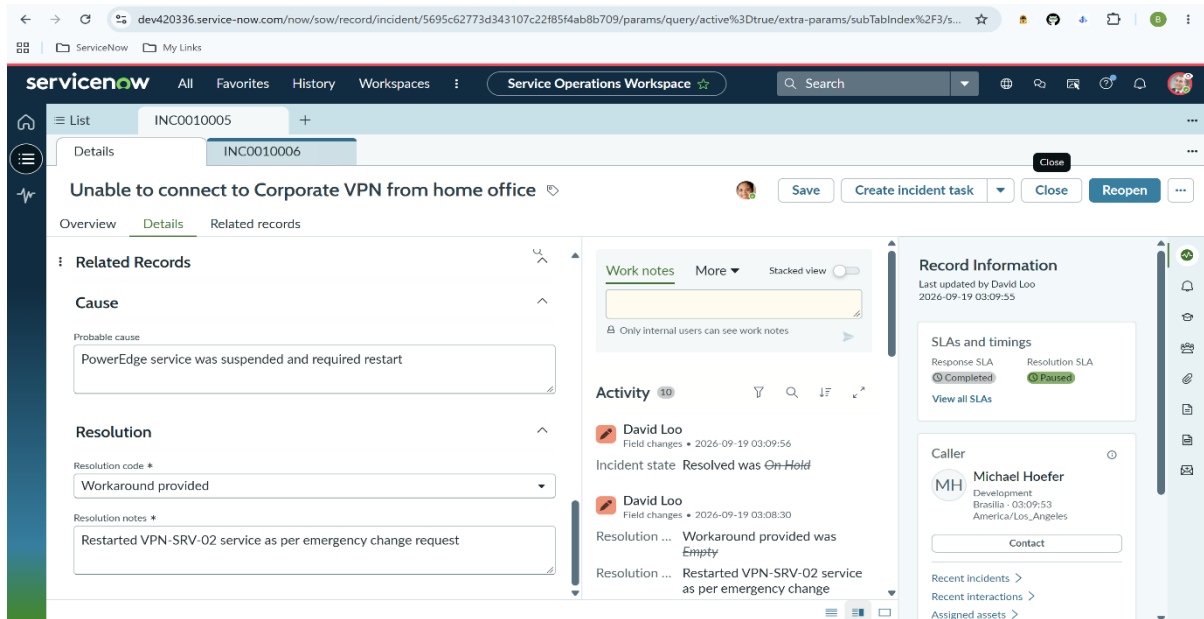


Figure 5.9: Cause and Resolution 1

5.12 Knowledge Article Creation

- After resolving the incident, select **Create Knowledge**.
- Select IT as Knowledge Base.
- Select article Template as Standard >> Next.
- Verify all the Details >> Save.
- Go back to Parent Record.
- Open Related Records → Verify Created Knowledge.
- Select **Next**.
- Verify the incident and resolution details.
- Save the Knowledge Article.
- Return to the parent incident and navigate to: **Related Records** → **Knowledge**
- Verify that the newly created Knowledge Article is linked to the incident.

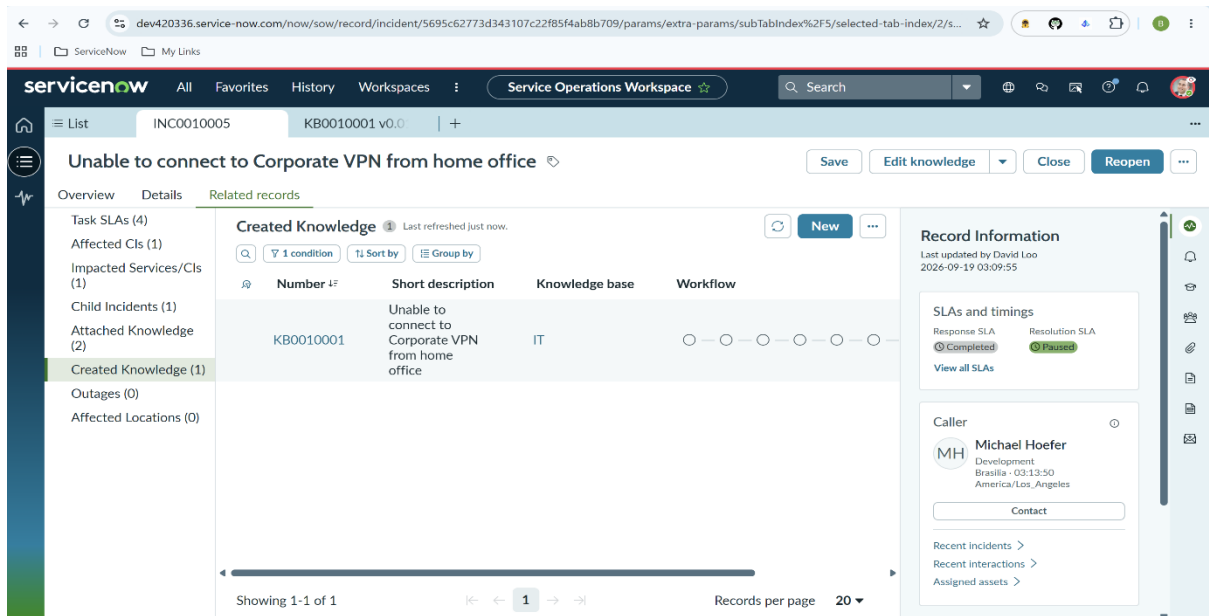


Figure 5.10: Knowledge Article Creation 1

5.13 Testing and Validation

Test Case	Expected Result
Incident Creation	Ticket created with caller and short description
Classification	Category, subcategory, urgency, service, offering, and CI populated correctly
Knowledge Integration	Relevant KB articles shown and attachable via Agent Assist
Reassignment	Ticket moved from Service Desk to Network
Level 2 Investigation	Full details, KB, and SLA visible to Level 2
Change Integration	Emergency Change created and linked
Child Incident	Created and linked to the parent
Resolution	Cause and resolution logged, ticket resolved
Knowledge Creation	KB article generated and linked back to incident
SLA Tracking	SLA status visible throughout via related records